

Name
Date
Course

Polynomials (Part II): Complex Zeros, Complex Zeros, & Finding Zeros of Polynomials

Find the zeros of the polynomial functions below and write each function in factored form.

① $f(x) = -4x^3 + 20x + 8$

② $f(x) = x^3 - x^2 + 2$

$$\textcircled{3} f(x) = 3x^3 - 28x + 3$$

$$\textcircled{4} f(x) = -x^3 - 2x^2 + 3$$

$$\textcircled{5} f(x) = x^4 - 2x^2 - 3$$

$$\textcircled{6} f(x) = 3x^4 + 17x^2 - 90$$

Find the zeros of the following polynomial functions using the grouping method.

⑦ $f(x) = 4x^3 + 2x^2 + 6x + 3$

⑧ $f(x) = 6x^3 - 18x^2 - 7x + 21$

$$\textcircled{9} f(x) = 9x^5 + 18x^4 - 4x - 8$$

$$\textcircled{10} f(x) = 3x^5 - 10x^4 - 75x + 250$$

$$\textcircled{11} f(x) = 5x^4 - x^3 - 320x + 64$$

$$\textcircled{12} f(x) = x^4 - x^3 - 15x + 15$$

Answers

① $-2, 1-\sqrt{2}, 1+\sqrt{2}$

$$f(x) = (x+2)(x-(1-\sqrt{2}))(x-(1+\sqrt{2}))$$

② $-1, 1-i, 1+i$

$$f(x) = (x+1)(x-(1-i))(x-(1+i))$$

③ $3, \frac{-9-\sqrt{93}}{6}, \frac{-9+\sqrt{93}}{6}$

$$f(x) = (x-3)(x-\frac{-9-\sqrt{93}}{6})(x-\frac{-9+\sqrt{93}}{6})$$

④ $1, \frac{-3-i\sqrt{3}}{2}, \frac{-3+i\sqrt{3}}{2}$

$$f(x) = (x-1)(x-\frac{-3-i\sqrt{3}}{2})(x-\frac{-3+i\sqrt{3}}{2})$$

⑤ $-\sqrt{3}, \sqrt{3}, -i, i$

$$f(x) = (x+\sqrt{3})(x-\sqrt{3})(x+i)(x-i)$$

⑥ $-\frac{\sqrt{30}}{3}, \frac{\sqrt{30}}{3}, 3i, -3i$

$$f(x) = (x+\frac{\sqrt{30}}{3})(x-\frac{\sqrt{30}}{3})(x-3i)(x+3i)$$

⑦ $\frac{1}{2}, -i\frac{\sqrt{6}}{2}, i\frac{\sqrt{6}}{2}$

$$f(x) = (x-\frac{1}{2})(x+i\frac{\sqrt{6}}{2})(x-i\frac{\sqrt{6}}{2})$$

$$\textcircled{8} \quad 3, -\frac{\sqrt{42}}{6}, \frac{\sqrt{42}}{6}$$

$$f(x) = (x-3)(x+\frac{\sqrt{42}}{6})(x-\frac{\sqrt{42}}{6})$$

$$\textcircled{9} \quad -2, -\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}, -i\frac{\sqrt{6}}{3}, i\frac{\sqrt{6}}{3}$$

$$f(x) = (x+2)(x+\frac{\sqrt{6}}{3})(x-\frac{\sqrt{6}}{3})(x+i\frac{\sqrt{6}}{3})(x-i\frac{\sqrt{6}}{3})$$

$$\textcircled{10} \quad \frac{10}{3}, -\sqrt{5}, \sqrt{5}, -i\sqrt{5}, i\sqrt{5}$$

$$f(x) = (x-\frac{10}{3})(x+i\sqrt{5})(x-i\sqrt{5})(x+i\sqrt{5})(x-i\sqrt{5})$$

$$\textcircled{11} \quad \frac{1}{5}, 4, -2-2i\sqrt{3}, -2+2i\sqrt{3}$$

$$f(x) = (x-\frac{1}{5})(x-4)(x-(-2-2i\sqrt{3}))(x-(-2+2i\sqrt{3}))$$

$$\textcircled{12} \quad 1, 5, \frac{-5-5i\sqrt{3}}{2}, \frac{5+5i\sqrt{3}}{2}$$

$$f(x) = (x-1)(x-5)(x-\frac{-5-5i\sqrt{3}}{2})(x-\frac{5+5i\sqrt{3}}{2})$$